Víctor García Carrasco

PhD in Statistics & Operations Research · Simulation & Optimisation · Energy Systems Modelling · Python

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Profile

Physicist with a PhD from UPC (2026, Excellent), specialising in the simulation and optimisation of complex systems. I build hourly simulation, discrete-event and multi-objective optimisation models in Python and publish them as web tools at orlabs.cloud. Applied experience in the electricity sector, the water-energy nexus and epidemiological simulation, with 7 publications. Looking to bring this profile to engineering, energy, data or R&D projects.

Technical skills

- **Programming:** Python (NumPy, pandas, SciPy, Numba, joblib, DEAP), automated batch pipelines with Excel and PDF output, web scraping, development and deployment of web applications.
- **Simulation:** discrete-event (GPSS), system dynamics (Insight Maker), SDL, cellular automata, digital twins, model validation.
- **Optimisation:** multi-objective, evolutionary algorithms, exact and heuristic methods for routing and networks (TSP/VRP).
- **Data analysis and predictive models:** multiple regression and autoregressive models, machine learning (sklearn) and neural networks (pytorch, tensorflow), time series, design of experiments, scenario analysis and visualisation.
- **Energy domain:** Iberian electricity mix, wholesale prices, curtailment and storage.

Experience

Universitat Politècnica de Catalunya (UPC) — Lecturer · 2023–present

Department of Statistics and Operations Research.

- Teach Simulation (BSc in Computer Engineering) and Discrete Network Models (MSc in Statistics and Operations Research).
- Design lab assignments on discrete-event simulation, design of experiments and system dynamics.

UPC — Doctoral researcher, water-energy nexus in Catalonia · 2021–2026

- Predictive model of the water volume stored in the Catalan internal basins.
- Hourly simulation model integrating renewable generation, electricity demand, reservoirs and desalination.
- Multi-objective optimisation of energy-transition scenarios: emissions, cost and water security.
- Opportunistic desalination strategy driven by renewable energy availability.
- Conference paper published by Springer (DSA ISC 2025).
- Thesis graded Excellent.

UPC — Researcher, COVID-19 epidemiological simulation · 2020–2023

- SDL-PAND project (UPC Centre for Development Cooperation): SEIRD models, digital twin and cellular automata to assess containment strategies in Catalonia.
- Co-author of 5 journal articles and 1 IEEE conference paper.

Personal projects

- **OR Labs** — operations research web platform that I designed, built and deployed, available in three languages with student access. Includes discrete-event simulation (PyGPSS-X), vehicle routing optimisation (Opt-Lab), hourly electricity mix analysis (EnerSim) and agent-based macroeconomic simulation (EcoSim).
- **Spanish electricity market** — work used in the preparation of the Greenpeace report Cierre nuclear y transición energética: el caso de Almaraz (nuclear phase-out and the energy transition). Included a wholesale price model driven by gas prices, with battery damping calibrated on CAISO data, applied to nuclear closure scenarios (Almaraz I and II).

Education

Year	Degree	Institution
2026	PhD — *The water-energy nexus in Catalonia facing the energy transition* (Excellent)	UPC
2012	Postgraduate course in Quantitative Techniques for Financial Markets	UPC
2008	MSc in Computational and Applied Physics	UPC

Year	Degree	Institution
2007	Degree in Physics (5-year Licenciatura)	UB

Publications

7 publications: 5 journal articles and 2 conference papers. Selected:

- García, V.; Fonseca, P. Energy Transition and Sustainable Water Management in Catalonia: Towards a Holistic Simulation Model. *Decision Sciences*, Springer, 2026.
- Fonseca, P.; García, J.; García, V. Modeling SARS-CoV-2 true infections in Catalonia through a digital twin. *Advanced Theory and Simulations*, 2023.
- Fonseca, P.; García, J.; García, V.; Pi, X.; Wainer, G. Formal modeling and simulation for SARS-CoV-2 containment scenarios in Catalonia. *Computing in Science & Engineering*, 2022.

Languages

Catalan and Spanish: native (C2) · English: reading B2, speaking and writing B1